

Geometric series



1 17

2 $\frac{5461}{64}$

3 9.143

4 683

5 20 475

6 a 14.29 b -204.92

7 a i $n = 10$ ii $S_{10} = -1364$ b i $n = 6$ ii $S_6 = \frac{1365}{16}$

8 a $n = 8$ b $n = 9$ c $n = 6$ d $n = 6$

9 a $\frac{1}{2}$ b 10.0

10 a $x \left(\frac{1 - \left(\frac{2}{5}x\right)^n}{1 - \frac{2}{5}x} \right)$ b 2555

11 $\frac{1 - 4 \times \left(\frac{1}{4}\right)^n + 3 \times \left(\frac{1}{5}\right)^n}{12}$

12 $r = 3$

13 $u_1 = 327\,680$

Summation notation



14

a

$$\sum_{k=1}^{10} 2^{k-1}$$

b

$$\sum_{k=1}^{11} \left(-\frac{1}{2}\right)^{k-1}$$

c

$$\sum_{k=1}^6 (-3)^k$$

d

$$\sum_{k=1}^5 6(3)^{k-1}$$

e

$$\sum_{k=4}^{10} \left(\frac{1}{2}\right)^k$$

f

$$\sum_{k=1}^7 16 \left(\frac{1}{2}\right)^k$$

15

a 340

b 7205

c 31

d 4914

e -252

f 242

g $\frac{3267}{4}$

h $-\frac{40}{27}$

16

$$\sum_{k=1}^{19} 2^{k-1} x^{3k-2}$$

Technology

17 5

18 8

19

a $u_1 r^4 = 405$

b $u_1 r^5 + u_1 r^6 + u_1 r^7 = 15\,795$

c $u_1 = 5, r = 3$

d $u_n = 5 \times 3^{n-1}$

Applications

20

a \$8191

b \$536 870 911

21

a \$59 559.81

b \$105 598.06

22



23

a 6832 m^3

b 7168 m^3

24 $n = 5$

25

a $20 \left(\frac{1 - 0.8^n}{0.2} \right) \text{ m}$

b 59.04 m

c 9 m

26

a 0.04

b $5000 \times 1.04 + 700$

c $5000 \times (1.04)^2 + 700 \times 1.04 + 700$

d $5000 \times (1.04)^3 + 700 \times (1.04)^2 + 700 \times 1.04 + 700$

e $5000 \times (1.04)^n + \frac{700((1.04)^n - 1)}{1.04 - 1}$

f $\$13\,293$

27

a

i $50\,000 \times 1.059 + 3000$

ii $50\,000 \times (1.059)^2 + 3000 \times 1.059 + 3000$

iii $50\,000 \times (1.059)^3 + 3000 \times (1.059)^2 + 3000 \times 1.059 + 3000$

b $\frac{3000((1.059)^n - 1)}{0.059}$

c $\$99\,791$

28

a $A_1 = 30\,000 \times 1.015 - 600$

b

i $A_2 = 30\,000 \times (1.015)^2 - (1.015 \times 600) - 600$

ii $A_3 = 30\,000 \times (1.015)^3 - (1.015^2 \times 600) - (1.015 \times 600) - 600$

c $A_n = 30\,000 \times (1.015)^n - \frac{600((1.015)^n - 1)}{1.015 - 1}$

d $\$22\,908.60$

29

a $1400 \times (1.005)^n + \frac{165((1.005)^n - 1)}{(1.005) - 1}$

b $\$8165.81$



30

a $A_1 = 190\,000 \times 1.07 - x$

b

i $A_2 = 190\,000 \times (1.07)^2 - 1.07x - x$

ii $A_3 = 190\,000 \times (1.07)^3 - (1.07)^2 x - 1.07x - x$

c $190\,000 \times (1.07)^n - \frac{x((1.07)^n - 1)}{0.07}$

d \$17 934.66

31

a $110\,000 \times 1.04^n - \frac{x(1.04^n - 1)}{0.04}$

b $x = 4488.88$

32

a $190\,000 \times 1.0025^n - \frac{x(1.0025^n - 1)}{0.0025}$

b $x = 801.05$